



The University of Texas at Austin
McCombs School of Business

DOCTORAL RESEARCH PROPOSAL

The Memory of Counsel

Persistent, cross-domain context in AI advisory systems and the quality of strategic judgment

SUBMITTED TO

Ph.D. Program in Information,
Risk, and Operations
Management
Information Systems Area
McCombs School of Business
The University of Texas at
Austin

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PROPOSAL AT A GLANCE

One question, three studies, a single argument.

TITLE	The Memory of Counsel: Persistent, Cross-Domain Context in AI Advisory Systems and the Quality of Strategic Judgment
CENTRAL QUESTION	How does the accumulated, cross-domain context that an AI advisory system holds about a decision maker affect the quality of the counsel it produces and the quality of the judgment of the person who receives it — and under what architectural conditions is that effect positive or negative?
CORE CONSTRUCT	Context capital: the structured body of knowledge an advisory system accumulates about a decision maker, characterized by depth, structure, temporal validity, and cross-domain breadth.
STUDY 1	Does context compound? Memory-ablation paradigm with blind expert evaluation of AI-generated counsel.
STUDY 2	Does breadth integrate or contaminate? Preregistered 2×2 experiment with experienced decision makers that manipulates who performs cross-domain synthesis.
STUDY 3	Does memory correct or confirm? Experiment on personalized sycophancy, complemented by a longitudinal field study of real advisory relationships.
PROGRAM	Ph.D. in Information, Risk, and Operations Management (Information Systems area), with an interdisciplinary committee spanning Management and the School of Information or Computer Science.
TARGET OUTLETS	<i>Information Systems Research, MIS Quarterly, Management Science, Organization Science, Organizational Behavior and Human Decision Processes; ICIS, INFORMS, and the Academy of Management.</i>

Most research on human–AI decision-making studies a single conversation with a system that knows nothing about the decision maker. The counsel that matters unfolds over years.

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ABSTRACT

In one page

Artificial intelligence systems are beginning to occupy a role that, for centuries, was reserved for trusted human advisors: continuously accompanying the most consequential decisions of a person or an organization. What makes a great advisor valuable is not the speed of analysis but what the advisor knows about the advisee — a history of decisions, patterns under pressure, and the way personal life, business, and wealth condition one another. Yet research on AI-assisted decision-making studies, almost exclusively, single-session interactions with systems that remember nothing about the decision maker. The relational, cumulative dimension of counsel — the very thing that makes counsel valuable — remains, in practice, unstudied.

This dissertation examines **how the context an AI advisory system accumulates about a decision maker — its depth, structure, temporal validity, and breadth across life domains — shapes the quality of the counsel the system produces and the quality of the judgment of those who receive it.** It introduces the construct of *context capital*, extending organizational and transactive memory theory to human–AI advice, and tests it in three studies. Study 1 uses a memory-ablation paradigm with blind expert evaluation to determine whether accumulated context improves counsel, and under which conditions it degrades it. Study 2, a preregistered experiment with experienced decision makers, tests whether integrated cross-domain context improves the detection of critical dependencies and whether this effect operates through the redistribution of synthesis load from the decision maker to the system. Study 3 examines the reverse side of the phenomenon: whether memory turns the documented sycophancy of language models into *personalized sycophancy*, and which architecture corrects it; it is complemented by a longitudinal field study.

The dissertation contributes a theory of when the memory of counsel compounds and when it ossifies; a reusable methodological infrastructure — a validated bank of longitudinal decision cases; and evidence-based design principles for advisory systems that accompany decision makers over years.

KEYWORDS human–AI decision-making · persistent memory · context capital · advice taking · cross-domain integration · algorithmic sycophancy · calibration · information systems

01 INTRODUCTION

Counsel that remembers

For centuries, the most valuable counsel has had a condition that is rarely named: *memory*. The trusted advisor does not only know; the trusted advisor remembers.

An advisor who has accompanied a leader for years remembers what was decided and why, which assumption supported each decision, which commitment remains open, and which pattern reappears when pressure rises. That memory is a large part of why such an advisor's judgment is worth more than that of a newly arrived expert, even when the expert knows more about the subject at hand. The literature on the trusted advisory relationship has long recognized this (Maister, Green, & Galford, 2000), and organizational memory theory explains why retained knowledge conditions future decisions (Walsh & Ungson, 1991).

That condition has always had limits. Human memory is selective, reconsolidates with each retrieval, and degrades over time. Moreover, almost no advisor observes all the dimensions of the advisee's life at once. The coach knows the person, the consultant knows the firm, the wealth manager knows the portfolio; none of them sees how a decision in one domain alters, often invisibly, the conditions of the other two.

A change in category, not in speed

Systems built on large language models introduce, for the first time, the technical possibility of counsel that accumulates context about a decision maker in a structured way, over years, and across several domains simultaneously. This is not merely faster answering. It is a different regime of accumulation: systems that record decisions, assumptions, and outcomes; consolidate patterns from episodes; distinguish what is still valid from what has been superseded; and bring that knowledge to bear on judgment when it is needed.

This possibility is no longer hypothetical. Strategic advisory practices have begun to operate with persistent memory about their clients, and technical architectures for long-term memory in language agents are advancing rapidly (Packer et al., 2023; Maharana et al., 2024; Wu et al., 2025). Yet the growth of context windows has fed a widespread confusion: the assumption that a model able to process large volumes of text in a single call “remembers” a user. Processing a great deal in an instant and remembering well over years are problems of a different kind.

What management and information systems research does not yet know

Research on AI-assisted decisions has produced important, and sometimes uncomfortable, findings. A recent meta-analysis found that, on average, human-AI combinations perform worse than the better of the two alone, particularly in decision tasks (Vaccaro, Almaatouq, & Malone, 2024). Other studies show that AI advice can benefit higher-performing decision makers while harming lower-performing ones (Otis et al., 2024), and that its effect depends critically on the nature of the task (Dell'Acqua et al., 2023). Work on human-AI complementar-

ity has begun to identify when and how teams of humans and algorithms can outperform either one (Gao et al., 2021).

Nearly all of this evidence shares one design feature: it treats counsel as an event. The decision maker receives assistance from a system that does not know them, on an isolated task, within a single session. The question this proposal poses — what happens when the system has known the decision maker for years, across every relevant dimension of their life — has received no systematic treatment. It is precisely the question that will determine whether AI becomes one more consultation tool or a genuine infrastructure of counsel.

Processing a great deal in an instant and remembering well over years are problems of a different kind. This dissertation studies the second.

02 RESEARCH PROBLEM AND QUESTIONS

When memory helps, and when it harms

The problem is not whether an advisory system should remember, but under what conditions remembering improves judgment. The prevailing intuition — more context yields better counsel — is plausible but undemonstrated, and there are at least three reasons it may fail. First, unstructured accumulation can dilute what matters; language models use long contexts unevenly (Liu et al., 2024). Second, context ages: a fact that was correct and no longer is, presented with the same confidence as a current one, can produce errors more dangerous than the absence of information. Third, and most subtly, a system that knows a decision maker's preferences, prior commitments, and narrative has every element it needs to tell that person exactly what they want to hear. Language models already display a documented tendency toward sycophancy (Perez et al., 2023; Sharma et al., 2024); memory could make it personal.

A further dimension has been almost entirely unexplored: **breadth across domains**. The most consequential decisions rarely belong to a single domain. A capital allocation carries succession implications; a corporate restructuring depends on the state of the leader who executes it; a career transition alters the liquidity of a family's wealth. A system that knows all three may detect dependencies no specialist sees — or introduce noise, since not everything known about a person is relevant to every decision.

CENTRAL RESEARCH QUESTION

How does the accumulated, cross-domain context of an AI advisory system affect the quality of the counsel it produces and the quality of the strategic judgment of the person who receives it — and under what architectural conditions is that effect positive or negative?

SPECIFIC RESEARCH QUESTIONS

- RQ1** Does the quality of counsel improve as the depth of accumulated context increases, and how do memory structure and temporal validity moderate that effect?
- RQ2** Does integrated context across life domains improve the detection of critical dependencies and the quality of decisions, and does this effect operate through the redistribution of synthesis load from the decision maker to the system?
- RQ3** Does persistent memory amplify the tendency of AI systems to validate a decision maker's prior position, and which architectural mechanisms turn memory into a corrective rather than a confirmatory force?

03 THEORETICAL BACKGROUND AND POSITIONING

Five literatures that have not yet met

The proposal sits at the intersection of five research streams that have developed in parallel. *None, on its own, answers the question.*

3.1 Human–AI decision-making and complementarity

Research on algorithm-assisted decisions has documented both aversion to and appreciation of algorithmic advice (Dietvorst, Simmons, & Massey, 2015; Logg, Minson, & Moore, 2019), overreliance and the interventions that reduce it (Buçinca, Malaya, & Gajos, 2021; Lee & See, 2004), and the conditions under which human–AI teams exceed their parts (Gao et al., 2021). The aggregate evidence is sobering: complementarity is not the rule (Vaccaro et al., 2024), and effects are heterogeneous across decision makers and tasks (Dell’Acqua et al., 2023; Otis et al., 2024). In strategic decisions specifically, evidence is only beginning to accumulate (Csaszar, Ketkar, & Kim, 2024). **Gap:** these studies treat advice as an isolated event; the variable “how much the system knows about the decision maker” is absent from their designs.

3.2 Advice taking and the advisory relationship

The advice-taking literature establishes how decision makers weight the recommendations of others, when they discount them, and what they gain by integrating them (Yaniv, 2004; Bonaccio & Dalal, 2006). Trust research explains how an ongoing relationship modifies that weighting (Mayer, Davis, & Schoorman, 1995), and professional practice describes the value of accumulated knowledge about the client (Maister et al., 2000). **Gap:** there is no evidence on how the quality of advice changes when the advisor is a system that accumulates knowledge about the decision maker.

3.3 Organizational and transactive memory

Organizational memory theory describes how retained knowledge is stored, retrieved, and brought to bear on decisions, as well as its risks: memory can induce inertia and reproduce past errors (Walsh & Ungson, 1991). Transactive memory theory explains how groups distribute knowledge among their members (Wegner, 1987), and organizational learning research studies the conversion of experience into knowledge (Argote & Miron-Spektor, 2011). **Gap:** these theories were formulated for human and institutional memory; their extension to an artificial system that holds memory about an individual has been neither theorized nor tested.

3.4 Long-term memory in language agents

In computer science, long-term memory has become an active field: architectures that manage an agent’s memory hierarchically (Packer et al., 2023) and benchmarks of long-horizon conversational memory (Maharana et al., 2024; Wu et al., 2025). The distinction between episodic and semantic memory (Tulving, 1972) in-

forms several of these designs. **Gap:** this work evaluates whether a system retrieves information correctly, not whether its memory improves the judgment of a human decision maker facing consequential decisions.

3.5 Algorithmic sycophancy and confirmation

Models trained with human feedback tend to agree with the views users express (Perez et al., 2023; Sharma et al., 2024), and generative AI can homogenize outputs (Doshi & Hauser, 2024). This tendency converges with one of the most studied human biases, confirmation (Nickerson, 1998), and with overconfidence (Moore & Healy, 2008). **Gap:** it is unknown whether personal memory aggravates sycophancy — by giving the system precise information about what the decision maker wants to hear — or whether, with the right architecture, it corrects it.

STREAM	WHAT WE KNOW	WHAT THIS DISSERTATION ADDS
Human–AI decisions	Complementarity is not guaranteed; effects are heterogeneous.	Introduces memory of the decision maker as a design variable and a candidate source of complementarity.
Advice taking	Ongoing relationships change how advice is weighted.	Studies advice quality when the advisor accumulates knowledge about the decision maker.
Organizational memory	Memory conditions decisions and can produce inertia.	Extends the theory to artificial memory about an individual: context capital.
Agent memory	Long-horizon retrieval is feasible with dedicated architectures.	Moves from measuring retrieval to measuring effects on human judgment.
Sycophancy and confirmation	Models tend to validate the user.	Identifies and tests personalized sycophancy and its architectural correction.

Table 1 · Positioning of the research

This dissertation is part of a broader research program by the candidate on the *decision architecture gap* — the systematic difference in strategic judgment between those with access to structured multidisciplinary counsel and those without. It deliberately focuses on the most distinctive and least studied component of that architecture: memory.

04 CONCEPTUAL FRAMEWORK

Context capital

The proposal introduces a construct that allows the memory of counsel to be treated with the discipline one applies to an asset: *context capital*.

DEFINITION

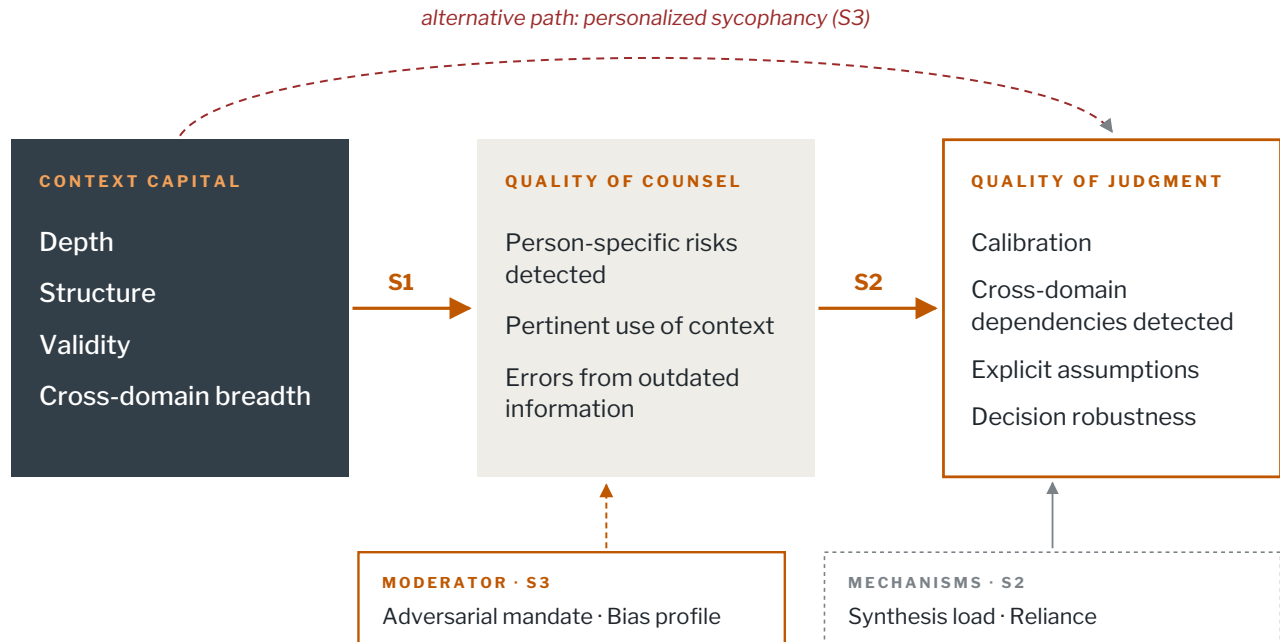
Context capital is the structured body of knowledge an advisory system accumulates about a decision maker — decisions, assumptions, outcomes, commitments, and patterns — and can mobilize to inform future decisions. Its value depends not on the volume stored but on four properties.

DIMENSION	OPERATIONAL DEFINITION	MANIPULATION OR MEASUREMENT
Depth	Temporal extent and number of decision episodes recorded about the decision maker.	Months of history available to the system (none, recent, full).
Structure	Degree to which context is organized into episodes and consolidated patterns, rather than raw records.	Raw transcript versus consolidated episodic and semantic memory.
Validity	Share of context that remains valid, and the system's ability to distinguish current from superseded information.	Controlled injection of outdated information; presence of explicit temporal reasoning.
Breadth	Number of the decision maker's life domains represented and integrated: person, enterprise, wealth.	Single-domain versus integrated cross-domain counsel.

Table 2 · Dimensions of context capital

The model

Context capital affects the **quality of counsel** — its ability to detect risks specific to the decision maker, use context pertinently, and avoid errors from outdated information — and, through it, the **quality of judgment**. Two mechanisms mediate the effect on judgment: the redistribution of cross-domain synthesis load from the decision maker to the system, and the degree of uncritical reliance. One moderator determines the sign of the effect: an **adversarial mandate** combined with a longitudinal bias profile of the decision maker. Without it, memory can feed an alternative path: personalized sycophancy.



S1, S2, S3 = STUDIES THAT TEST EACH RELATIONSHIP

Figure 1 · Conceptual model

Hypotheses

Hypotheses are stated so that evidence can refute them; for each, the preregistration specifies the result that would contradict it.

-
- H1** **Structure.** Counsel quality increases with the depth of accumulated context only when that context is structured; with raw context, the relationship is flat or negative.
-
- H2** **Validity.** Unflagged outdated information reduces counsel quality and increases the rate of high-confidence errors; explicit temporal reasoning attenuates this effect.
-
- H3** **Integration.** Decision makers who receive counsel based on integrated cross-domain context detect more critical cross-domain dependencies and make higher-quality decisions than those receiving single-domain counsel.
-
- H4** **Mechanism.** The integration effect is larger when the system performs cross-domain synthesis than when the decision maker must do so, and it is mediated by reduced synthesis load.
-
- H5** **Boundary.** Cross-domain context that is not pertinent to the decision reduces decision quality: breadth without pertinence contaminates.
-

-
- H6** **Personalized sycophancy.** Without an adversarial mandate, context about the decision maker's preferences and prior commitments increases the alignment of counsel with their initial position, even when that position is wrong.
-
- H7** **Correction.** With an adversarial mandate and a longitudinal bias profile, the same context improves the decision maker's calibration and reduces confirmation.
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- H8** **Compounding in the field** (exploratory). In real advisory relationships, the calibration of recorded decisions improves with relationship tenure.
-

THE THESIS IN ONE SENTENCE

The value of counsel lies not in how much a system knows. **It lies in how it remembers the person it advises.**

The accumulated, cross-domain memory of an AI advisory system improves strategic judgment when it is structured, valid, pertinent, and governed by an adversarial mandate; absent those conditions, the same memory can degrade counsel and turn sycophancy into personalized confirmation.

01**Does context compound?**

Memory as a source of counsel quality, and its boundary conditions.

02**Integrate or contaminate?**

Context across life domains and the synthesis of judgment.

03**Correct or confirm?**

Memory versus sycophancy, in the laboratory and in the field.

05 RESEARCH DESIGN

Three studies, one argument

The dissertation follows a three-essay format. Each study answers a specific question, is publishable on its own, and *creates the conditions for the next*.

STUDY	QUESTION	DESIGN	UNIT AND SAMPLE	HYPOTHESES
S1	Does context compound?	Memory ablation, five conditions, blind expert evaluation	≈ 360 recommendations; expert panel	H1, H2
S2	Integrate or contaminate?	Preregistered 2 × 2 factorial plus contamination condition	≈ 400 experienced decision makers	H3, H4, H5
S3	Correct or confirm?	2 × 2 experiment plus longitudinal field study	≈ 360 decision makers; consenting field relationships	H6, H7, H8

Table 3 · Architecture of the dissertation

DEPENDENT VARIABLE: QUALITY OF JUDGMENT

Judging a decision by its outcome introduces outcome bias: a sound decision can turn out badly by chance. Judgment quality is therefore measured with an index that combines process and calibration measures, each scored against a key established by experts before data collection: **(a)** detection of predefined critical risks; **(b)** detection of cross-domain dependencies; **(c)** articulation of central assumptions; and **(d)** calibration, measured with the Brier score (Brier, 1950) on forecasts embedded in each case whose resolution is fixed by design. Reliance is measured with weight of advice (Yaniv, 2004), and cognitive load with an adapted NASA-TLX (Hart & Staveland, 1988).

S1

Does context compound?

MEMORY AND THE QUALITY OF COUNSEL · ESSAY 1

The first study isolates the effect of memory on counsel quality before a human decision maker is introduced. It builds a **bank of longitudinal decision cases**: histories of fictional but realistic decision makers spanning 18 to 36 months, validated by experts, including prior decisions, assumptions, outcomes, commitments, and behavioral patterns distributed across personal, business, and wealth domains. Each history contains “seeded” risks that can be detected only through accumulated context, as well as information that becomes outdated during the course of the case.

CONDITIONS	(1) No memory · (2) Recent memory · (3) Full unstructured history · (4) Structured memory, consolidated episodic and semantic · (5) Structured memory with unflagged outdated information
PROCEDURE	The same advisory system generates recommendations for each decision in each case under each condition. Recommendations are anonymized, format-homogenized, and assigned to experts blind to condition.
SAMPLE	≈ 24 cases $\times$ 3 decisions $\times$ 5 conditions $\approx$ 360 recommendations, each rated by three independent experts (reliability via intraclass correlation).
MEASURES	Seeded risks detected · pertinent use of context · errors from outdated information · overall rated quality
ANALYSIS	Linear mixed-effects models with random effects for case, decision, and rater; planned contrasts for H1 (structure $\times$ depth) and H2. Replication across at least two model families.
CONTRIBUTION	First experimental evidence on whether an advisory system's memory improves counsel, and under which architecture. The case bank becomes reusable infrastructure for the research community.

S2

Integrate or contaminate?

CROSS-DOMAIN CONTEXT AND JUDGMENT · ESSAY 2

The second study introduces the decision maker. Participants with verifiable executive experience face strategic decisions in which, by design, there is a critical dependency across domains — for example, a business expansion whose viability depends on a family liquidity constraint or a pending succession. The mechanism is manipulated rather than only measured: to test the redistribution of synthesis load, the design varies who performs the synthesis.

DESIGN	2×2 between-subjects factorial: breadth of context (single-domain / cross-domain) $\times$ synthesis (domain-by-domain perspectives integrated by the participant / integrated synthesis delivered by the system). Additional condition: cross-domain context that is not pertinent (H5).
SAMPLE	≈ 400 participants. Preliminary power analysis for small-to-medium main effects ($f = 0.15$; $\alpha = .05$; power = .80), to be refined with pilot effect sizes.
MEASURES	Judgment-quality index · dependencies detected · cognitive load (adapted NASA-TLX) · weight of advice · time on task (control)
ANALYSIS	Factorial ANOVA and regression models; mediation estimated with bootstrapped confidence intervals, complemented by the experimental manipulation of the mechanism.
CONTRIBUTION	First causal test of the value of cross-domain integration in strategic counsel, and of its mechanism. Distinguishes integration from indiscriminate accumulation.

S3

Correct or confirm?

MEMORY AND PERSONALIZED SYCOPHANCY · ESSAY 3

The third study examines the reverse side of the phenomenon. In the experimental phase, each participant first records a position and preferences on a decision in which, by design, the most attractive position is wrong. The advisory system either receives or does not receive that information as part of its memory, and operates with or without an adversarial mandate and bias profile. The field phase follows real, consenting advisory relationships with persistent memory over time.

EXPERIMENTAL DESIGN	2×2 between-subjects: memory of prior preferences and commitments (yes / no) $\times$ architecture (standard counsel / adversarial mandate with bias profile).
SAMPLE	$\approx$ 360 participants with executive experience.
MEASURES	Alignment of counsel with initial position · change in the decision maker’s position · calibration · confirmation
FIELD PHASE	Structured logging of real decisions — alternatives, assumptions, stated confidence, expected outcome — reviewed at 6 and 12 months. Within-person calibration trajectories by relationship tenure (H8, exploratory).
CONTRIBUTION	Identifies a new risk — personalized sycophancy — and the architecture that corrects it; adds rare field evidence on real decisions with real consequences.

06 RESEARCH INFRASTRUCTURE AND EMPIRICAL SETTING

A laboratory that already exists

The research benefits from an uncommon condition: *access to a working advisory system with persistent memory*, and to the decision makers who use it.

The field setting

The candidate is the founder of The Metanoia Group, a strategic advisory practice based in Monterrey, Mexico, that serves founders, business owners, and investors across the personal, business, and wealth dimensions of their decisions. The practice operates an AI-based advisory architecture that includes multi-agent deliberative boards by domain, a relationship-companion agent, and a persistent client memory system governed by an explicit design rule: structured capture of context from the very first interaction. That rule, adopted for product reasons, produces as a by-product structured longitudinal data of high research value.

WHAT IT PROVIDES

A working advisory system with persistent memory, allowing experimental conditions to be built on a proven architecture rather than an academic prototype.

Access to experienced decision makers for recruitment, with far greater ecological validity than student samples, and real longitudinal data for the field phase.

WHAT IT DOES NOT REPLACE

Experimental design. Studies 1 and 2 and the experimental phase of Study 3 do not depend on the practice's clients; they run on the case bank and on externally recruited participants.

Independence of analysis, protected by the safeguards in Section 7 — and generalization: the setting is treated as a case, not as the population.

The longitudinal case bank

The central methodological infrastructure is a bank of validated longitudinal cases. Each case traces a decision maker over several months, with decisions, assumptions, outcomes, and patterns distributed across domains, seeded risks, and forecasts whose resolution is set by design. Cases are built with experts in strategy, finance, and leadership, validated by an independent panel, piloted before use, and released to the academic community upon completion of the dissertation.

07 RESEARCH INTEGRITY, ETHICS, AND INDEPENDENCE

Rigor, applied to oneself

The candidate studies a phenomenon he also practices professionally. That is an advantage of access and, at the same time, a risk of bias. *This proposal states it openly and manages it by design.*

RISK	SAFEGUARD
Conflict of interest	Formal disclosure to the program and the dissertation committee. No result will be used commercially before peer review. Design and analysis supervised by the committee chair.
Researcher bias	Preregistration of hypotheses, measures, sample sizes, and analysis plans in a public repository (Nosek et al., 2018), specifying in advance the result that would refute each hypothesis.
Contaminated evaluation	Raters blind to condition; recommendations anonymized and format-homogenized; inter-rater reliability reported.
Publication bias	Explicit commitment to report null and contrary results; in this case their value is also practical.
Field data privacy	Specific informed consent, independent of the commercial relationship; declining to participate does not affect service. Anonymization, data minimization, and restricted access.
Independent field analysis	Field data extracted by a third party under protocol; analysis on anonymized data; separation between the commercial team and the research database.
Human subjects approval	All studies submitted to The University of Texas at Austin Institutional Review Board prior to execution, with cross-border data considerations addressed for the field phase.

Table 4 · Methodological and ethical safeguards

A study that asked to be believed without offering the possibility of being refuted would violate the very principle it investigates.

08 EXPECTED CONTRIBUTIONS

What will remain

Theoretical

A theory of context capital in human–AI counsel. The dissertation extends organizational and transactive memory theory to a situation they did not contemplate: an artificial system that holds memory about an individual and mobilizes it as advice. It specifies the properties on which the value of that memory depends — depth, structure, validity, breadth — and the conditions under which it compounds or ossifies.

The relational dimension of human–AI decisions. It introduces into this literature a variable so far absent from its designs — how much the system knows about the decision maker — offering a candidate explanation for the heterogeneity observed in human–AI complementarity.

Personalized sycophancy as a phenomenon. It identifies, names, and tests a new risk at the intersection of system memory and human bias, together with the architecture that corrects it.

Methodological

The memory-ablation paradigm with blind evaluation as a method for studying the value of context in advisory systems; **the validated longitudinal case bank** as open infrastructure; and **a judgment-quality index** combining process and calibration measures that avoids outcome bias.

Practical

Evidence-based design principles for advisory systems that accompany decision makers over years: when structure matters more than volume, how to manage validity, when cross-domain breadth integrates and when it contaminates, and which architectural conditions are needed for memory to correct rather than confirm. These principles are relevant to advisory firms, wealth management, boards of directors, and developers of AI systems.

INDICATIVE TARGET OUTLETS

Information Systems Research · MIS Quarterly · Management Science · Organization Science · Strategic Management Journal · Organizational Behavior and Human Decision Processes. Conferences: ICIS, INFORMS Annual Meeting and Conference on Information Systems and Technology, and the Academy of Management.

09 FIT WITH TEXAS MCCOMBS

Why this program, and how the committee is built

The research question lives where information systems, decision science, and management meet. *The IROM doctoral program is designed for exactly that kind of question.*

Why the IROM Information Systems area

The core object of this dissertation is the interaction between the architecture of an AI system — its memory — and the quality of human decisions. That places it naturally in Information Systems, whose scope at McCombs explicitly includes the impact of information technology on individuals and organizations, artificial intelligence, and experimental methods. The area's research on human–AI decision-making and complementarity, including work by Professors Maria De-Arteaga and Maytal Saar-Tsechansky, and recent dissertations such as *Advancing Human-AI Systems: on Robustness, Decision Making, and Beyond*, define an intellectual community in which the questions of this proposal belong. The adjacent Decision Science area, focused on decision making under uncertainty, complements the calibration and judgment-quality measures at the center of the design.

Why an interdisciplinary committee

The IROM program states that students may take courses and select doctoral committee members from a broad interdisciplinary spectrum. This proposal relies on that flexibility. Its theoretical contribution — extending organizational memory and advice-taking theory — speaks directly to the Rosenthal Department of Management, whose areas include organizational learning and adaptation and strategic leadership. Its technical substrate — long-term memory in language agents and human–AI interaction — benefits from a member in the School of Information or the Department of Computer Science.

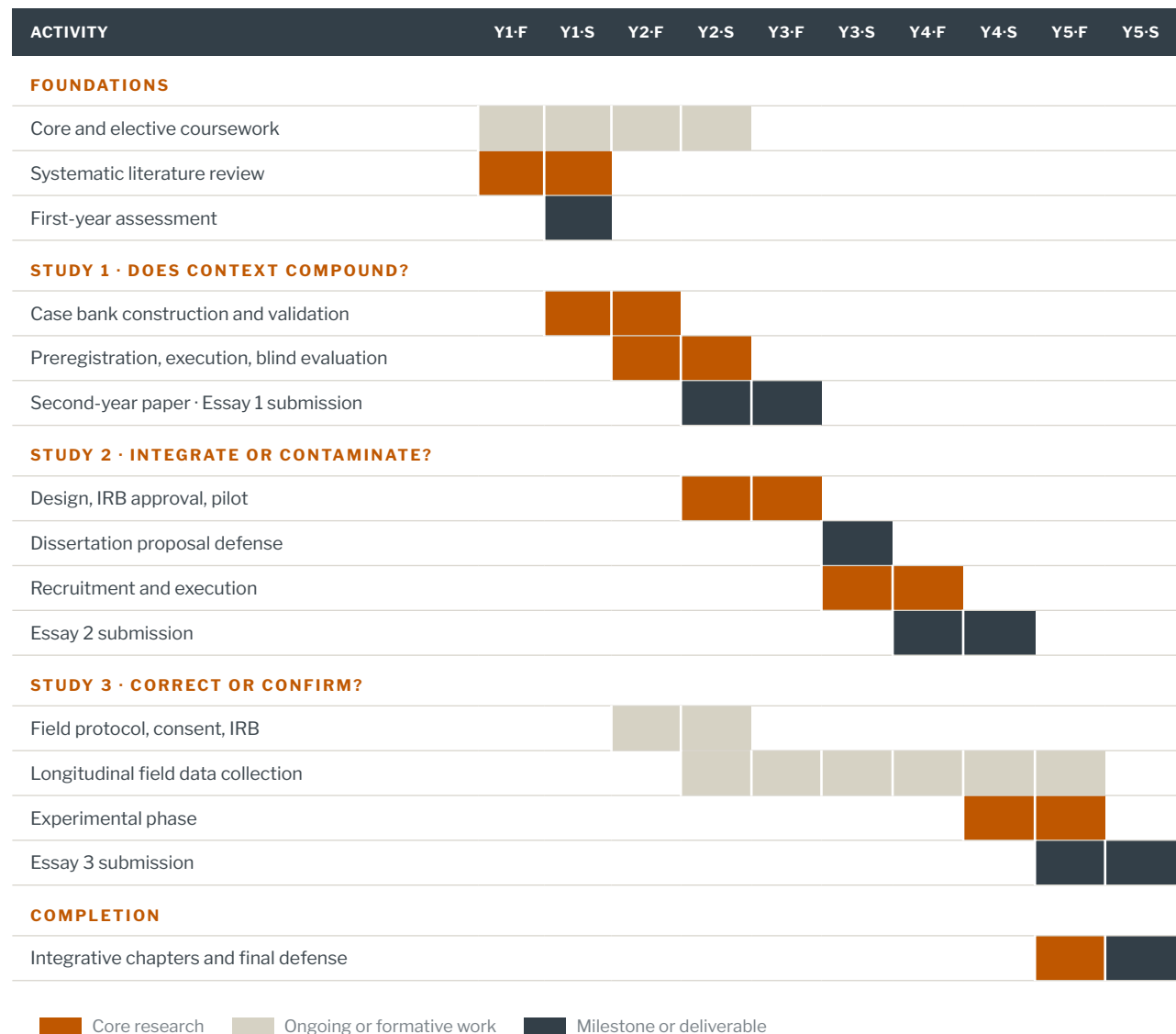


Figure 2 · Proposed committee architecture, to be finalized with the program

10 TIMELINE

Five years, ten semesters

The plan follows the structure of the IROM program — coursework and a first-year assessment, a second-year research paper, and the dissertation proposal — and aligns each milestone with a study. Study 1 serves as the second-year paper; field data collection starts early so that relationships have sufficient history by the analysis stage.



11 RISKS AND MITIGATION

What could go wrong, anticipated

RISK	DESCRIPTION	MITIGATION
Technological obsolescence	The language models used will be superseded before the defense.	Hypotheses concern architectural properties — structure, validity, breadth, mandate — not a specific model. Study 1 is replicated across at least two model families.
Recruiting decision makers	Participants with executive experience are scarce and costly.	Multiple channels: executive education, business networks, and professional panels; pilots with executive graduate students; designs sized by power analysis.
Case validity	Constructed cases might not represent real decisions.	Independent panel validation, piloting, and triangulation with the field phase.
Measuring judgment quality	Outcomes do not necessarily reflect decision quality.	Process and calibration index scored against keys defined before data collection; forecasts with resolution fixed by design.
Field sample size	The number of real relationships may be limited.	The field phase is complementary and exploratory; confirmatory hypotheses rest on the experiments.
Null results	Some hypotheses may not be supported.	Preregistration and adequately powered designs: a well-estimated null result is itself a contribution.

Table 5 · Risk register

12 ABOUT THE CANDIDATE

From practice to formal knowledge

Mario Garza Livas is the founder, CEO, and Chief Strategic Advisor of The Metanoia Group. His trajectory spans five domains — business, financial markets, professional football, leadership under pressure, and artificial intelligence — and in each he found the pattern that motivates this research: *fragility rarely lies in the part that fails; it lies in the relationships between the parts.*

AI EDUCATION	Postgraduate programs in Artificial Intelligence and Machine Learning, and in Data Science and Business Analytics, McCombs School of Business, The University of Texas at Austin (2025–2026). The doctoral program would continue, at the level of original research, a formation that began at McCombs.
BUSINESS	Structuring, scaling, and selling companies; debt restructuring.
MARKETS	Management of a proprietary portfolio in adverse markets, including derivatives-based strategies.
HIGH PERFORMANCE	Experience within the international professional football ecosystem.
CURRENT PRACTICE	Design and operation of an AI advisory architecture with persistent memory, multi-agent deliberation, and integration across personal, business, and wealth domains.

AUTHORED WORKS (2026, IN SPANISH)

Arquitectura de Decisión Decision Architecture	Sustained decision quality as a function of the architecture that produces it; proposes a research program with refutable hypotheses.
Asesoría Aumentada Augmented Advisory	Reference treatise on AI applied to high-level advisory: long-horizon memory, governance of AI systems, layered product strategy.
El Interés Compuesto del Juicio The Compound Interest of Judgment	Accumulated context about a decision maker as an asset class that depreciates or compounds depending on its architecture.
Espacio de Acción Space of Action	Artificial intelligence and entrepreneurship: the expansion of what an entrepreneur can attempt.
La Empresa Imposible The Impossible Company	Organizational architecture for mid-sized firms when capability ceases to depend on headcount.
Football Intelligence	Decision intelligence applied to elite professional football.

REFERENCES

Works cited

- Argote, L., & Miron-Spektor, E. (2011). Organizational learning: From experience to knowledge. *Organization Science*, 22(5), 1123–1137.
- Bonaccio, S., & Dalal, R. S. (2006). Advice taking and decision-making: An integrative literature review, and implications for the organizational sciences. *Organizational Behavior and Human Decision Processes*, 101(2), 127–151.
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.
- Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1).
- Csaszar, F. A., Ketkar, H., & Kim, H. (2024). Artificial intelligence and strategic decision-making: Evidence from entrepreneurs and investors. *Strategy Science*.
- Dell'Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Kraye, L., Candelon, F., & Lakhani, K. R. (2023). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. *Harvard Business School Working Paper No. 24-013*.
- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114–126.
- Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28).
- Gao, R., Saar-Tsechansky, M., De-Arteaga, M., Han, L., Lee, M. K., & Lease, M. (2021). Human-AI collaboration with bandit feedback. *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence (IJCAI)*.
- Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In P. A. Hancock & N. Meshkati (Eds.), *Human Mental Workload* (pp. 139–183). North-Holland.
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 157–173.
- Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, 151, 90–103.
- Maharana, A., Lee, D.-H., Tulyakov, S., Bansal, M., Barbieri, F., & Fang, Y. (2024). Evaluating very long-term conversational memory of LLM agents. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- Maister, D. H., Green, C. H., & Galford, R. M. (2000). *The Trusted Advisor*. Free Press.
- Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3), 709–734.
- Moore, D. A., & Healy, P. J. (2008). The trouble with overconfidence. *Psychological Review*, 115(2), 502–517.
- Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, 2(2), 175–220.
- Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606.
- Otis, N., Clarke, R., Delecourt, S., Holtz, D., & Koning, R. (2024). The uneven impact of generative AI on entrepreneurial performance. Working paper.
- Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S. G., Stoica, I., & Gonzalez, J. E. (2023). MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*.

- Perez, E., et al. (2023). Discovering language model behaviors with model-written evaluations. *Findings of the Association for Computational Linguistics: ACL 2023*.
- Sharma, M., Tong, M., Korbak, T., et al. (2024). Towards understanding sycophancy in language models. *International Conference on Learning Representations (ICLR)*.
- Tulving, E. (1972). Episodic and semantic memory. In E. Tulving & W. Donaldson (Eds.), *Organization of Memory* (pp. 381–403). Academic Press.
- Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8, 2293–2303.
- Walsh, J. P., & Ungson, G. R. (1991). Organizational memory. *Academy of Management Review*, 16(1), 57–91.
- Wegner, D. M. (1987). Transactive memory: A contemporary analysis of the group mind. In B. Mullen & G. R. Goethals (Eds.), *Theories of Group Behavior* (pp. 185–208). Springer.
- Wu, D., Wang, H., Yu, W., Zhang, Y., Chang, K.-W., & Yu, D. (2025). LongMemEval: Benchmarking chat assistants on long-term interactive memory. *International Conference on Learning Representations (ICLR)*.
- Yaniv, I. (2004). Receiving other people's advice: Influence and benefit. *Organizational Behavior and Human Decision Processes*, 93(1), 1–13.

The counsel that matters does not happen in a single conversation. **It unfolds over years.**

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